

|           |                                                                                                                                                                                                                                                                                                                             |          |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2(a)(i)   | <i>independent variable:</i> type / AW , of trypsin ;<br><i>dependent variable:</i> time (taken to break down gelatine) ;                                                                                                                                                                                                   | <b>2</b> |
| 2(a)(ii)  | any <b>one</b> from:<br><br>1    pH (of solution) ;<br>2    surface area / mass / volume, of cube ;<br>3    concentration of, gelatine / collagen, in cube ;                                                                                                                                                                | <b>1</b> |
| 2(b)(i)   | <i>mean</i> = 131.1 ;                                                                                                                                                                                                                                                                                                       | <b>1</b> |
| 2(b)(ii)  | <i>s</i> = 3.2 ;                                                                                                                                                                                                                                                                                                            | <b>1</b> |
| 2(b)(iii) | any <b>three</b> from:<br><br>1    critical value (at degrees of freedom = 22 and $p = 0.05$ ) = 2.074 ;<br>2    calculated ( $t$ -) value / 2.663 is, greater than, critical ( $t$ -) value / 2.074 ; <b>ora</b><br>3    null hypothesis is rejected ;<br>4    there is a <u>significant difference</u> (at $p = 0.05$ ) ; | <b>3</b> |

| Question | Answer                                                                                                                                                                                                                                                                                                 | Marks |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 2(b)(iv) | <p>any <b>two</b> from:</p> <ol style="list-style-type: none"> <li>1 <i>idea that</i> salmon trypsin is adapted to colder temperatures ; <b>ora</b></li> <li>2 (at 20 °C) salmon trypsin forms enzyme–substrate complexes at a faster rate (than pig trypsin) ; <b>ora</b></li> <li>3 AVP ;</li> </ol> | 2     |